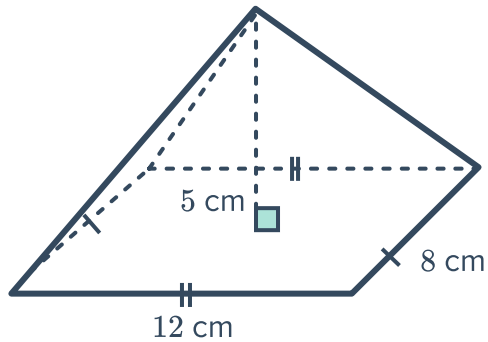


Volume of a pyramid

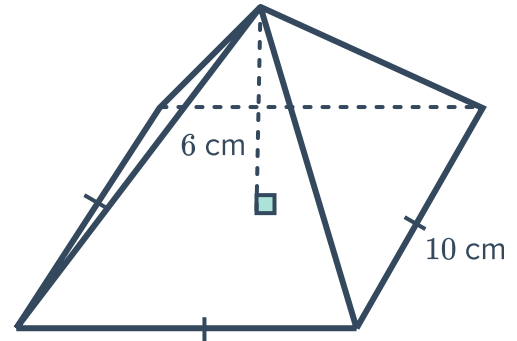


- 1 Find the volume of the following pyramids, rounding your answers to two decimal places where necessary:

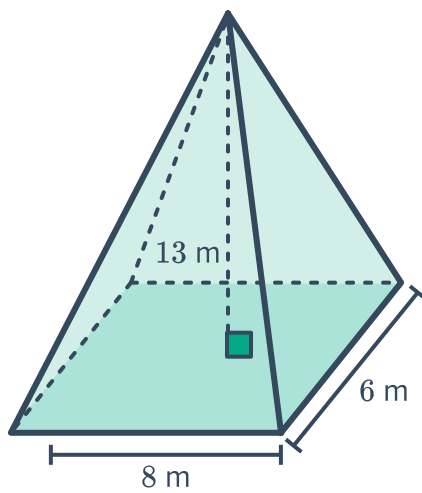
a



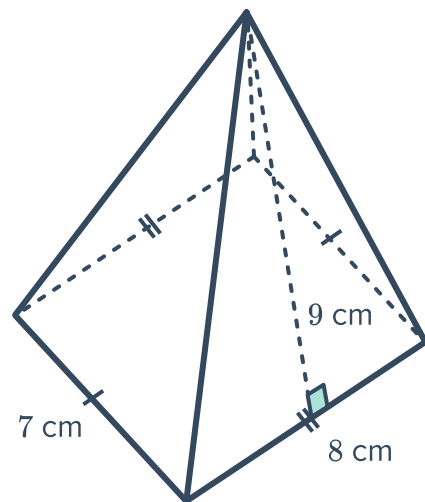
b



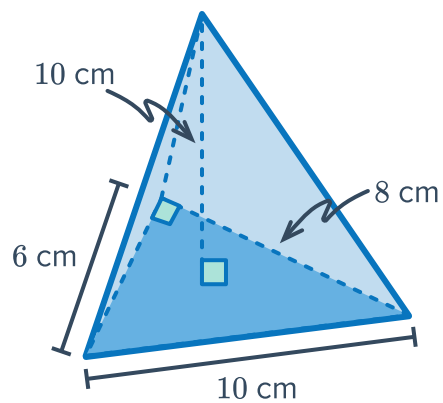
c



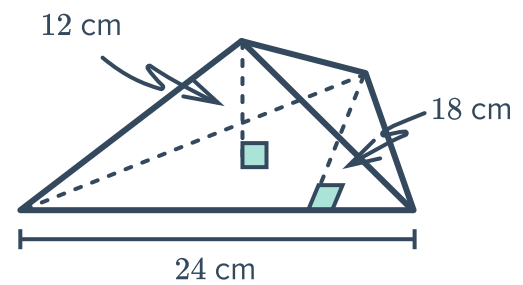
d



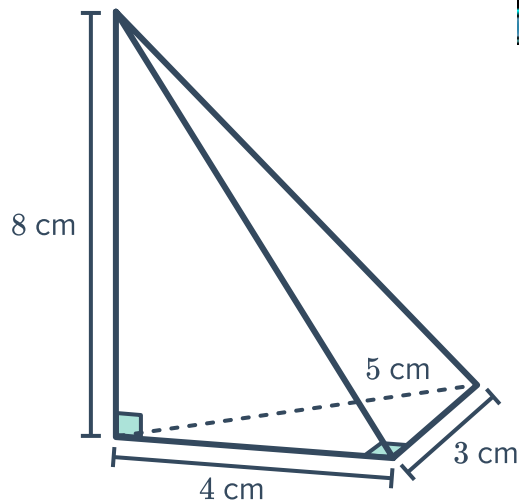
e



f

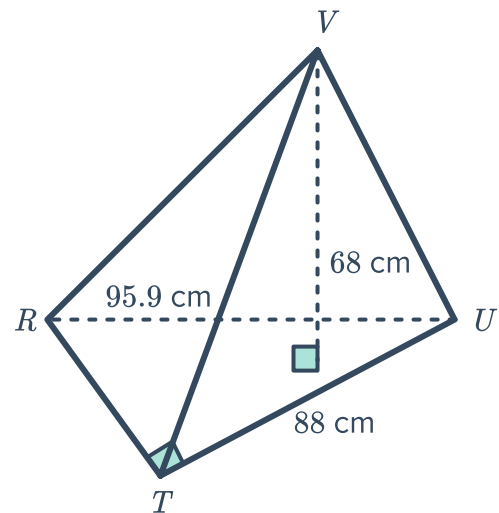


g



2 In the triangular pyramid shown:

- a Find the length of RT in centimetres, writing your answer correct to two decimal places.
- b Find the area of the base $\triangle RTU$.
- c Hence, find the volume of the pyramid correct to two decimal places.



3 Find the volume of the following pyramids:

- a A right pyramid with a square base of 24 m and a perpendicular height of 25 m.
- b A pyramid with a height of 5 m and a right-angled triangular base with side lengths 5 m, 12 m and 13 m.

4 Find the perpendicular height of the following pyramids:

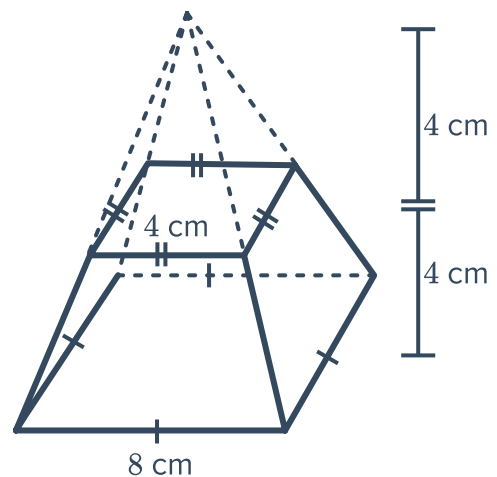
- a A right square pyramid with a base of length 24 cm and a volume 2304 cm^3 .
- b A rectangular pyramid with a volume of 234 cm^3 , a base width of 13 cm and base length of 9 cm.

5 Find the base length of the following pyramid 

- a A right square pyramid with a perpendicular height of 36 mm and a volume of 6912 mm^3 .
- b A rectangular pyramid with a volume of 180 cm^3 , a perpendicular height of 12 cm and a base width of 5 cm.

6 Find the volume of a pentagonal pyramid with base area of 90 cm^2 and height 8 cm.

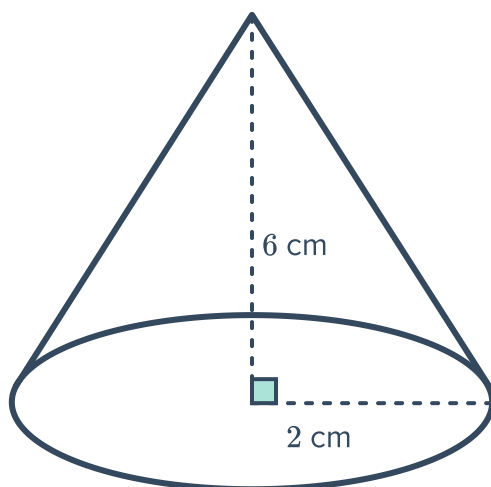
7 A small square pyramid of height 4 cm was removed from the top of a large square pyramid of height 8 cm forming the solid shown. Find the exact volume of the solid.



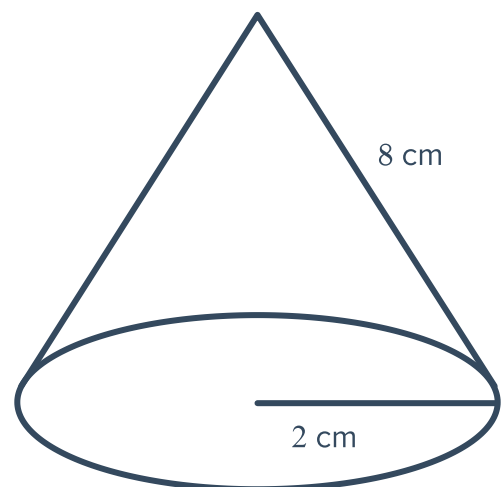
Volume of a cone

8 Find the volume of the cones shown below. Round your answer to two decimal places.

a



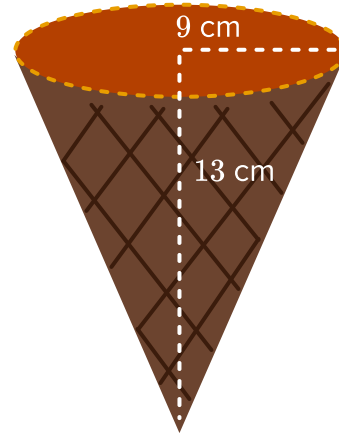
b



- 9 Find the capacity of the ice-cream cone in millilitres.

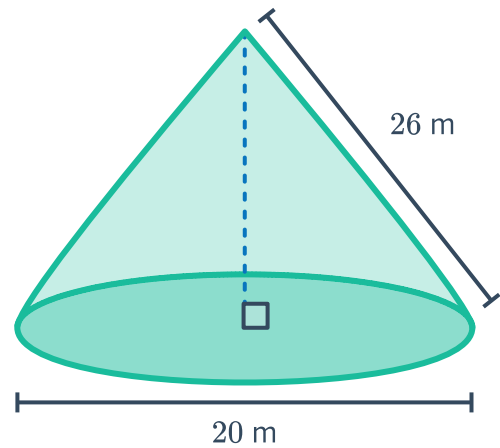


Round your answer to one decimal place.



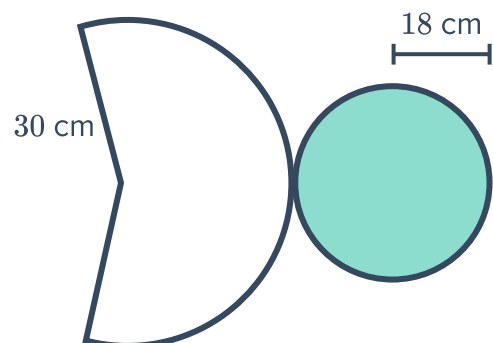
- 10 Consider the cone on the right:

- a Calculate the perpendicular height.
- b Hence find the volume of the cone.
Round your answer to three significant figures.

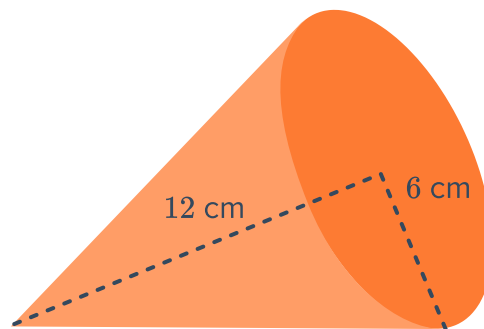


- 11 Consider the following net of a cone:

- a Which line segment has a length of 18 cm?
- b Which line segment has a length of 30 cm?
- c Find the perpendicular height, h , of the cone.
- d Find the volume of the cone in cubic centimetres. Round your answer to the nearest cubic centimetre.



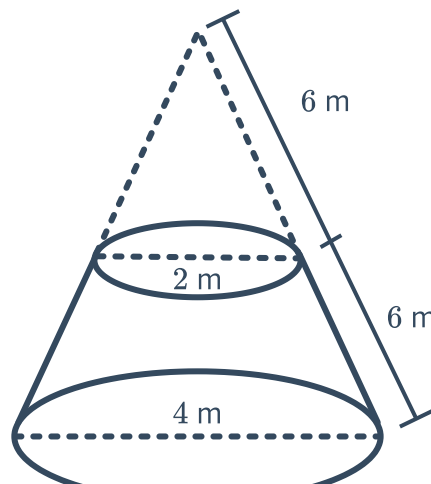
- 12 Find the volume of the cone shown. Round your answer to one decimal place.



- 13 Find the volume of a cone that has a radius of 8 cm and a perpendicular height 16 cm. Round your answer to two decimal places.
- 14 A cone has a volume of 160 mm^3 . If the perpendicular height and radius of the cone are equal in length, find the radius correct to two decimal places.
- 15 Find the radius of a cone that has a volume of 2456.97 cm^3 and a perpendicular height of 23 cm. Round your answer to one decimal place.
- 16 A cone has a volume of 1696.46 mm^3 and a perpendicular height of 20 mm. Find the radius of the cone. Round your answer to the nearest mm.
- 17 A cone has a volume of $106\,532.52 \text{ cm}^3$ and a radius of 54.7 cm. Find the height of the cone. Round your answer to the nearest cm.

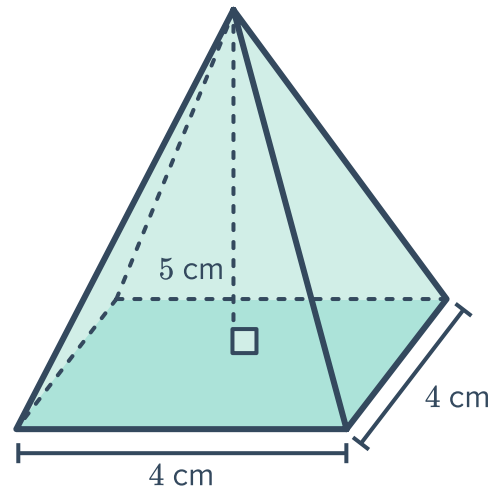
Applications

- 18 The following podium was formed by sawing off the top of a cone. Find its volume, correct to two decimal places.



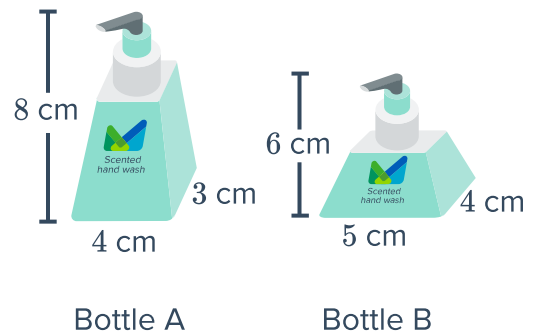


- 19** A paperweight is in the shape of a square pyramid with dimensions as shown. The paperweight is filled with solid glass.
- Find the volume of glass needed to make 3000 paperweights.

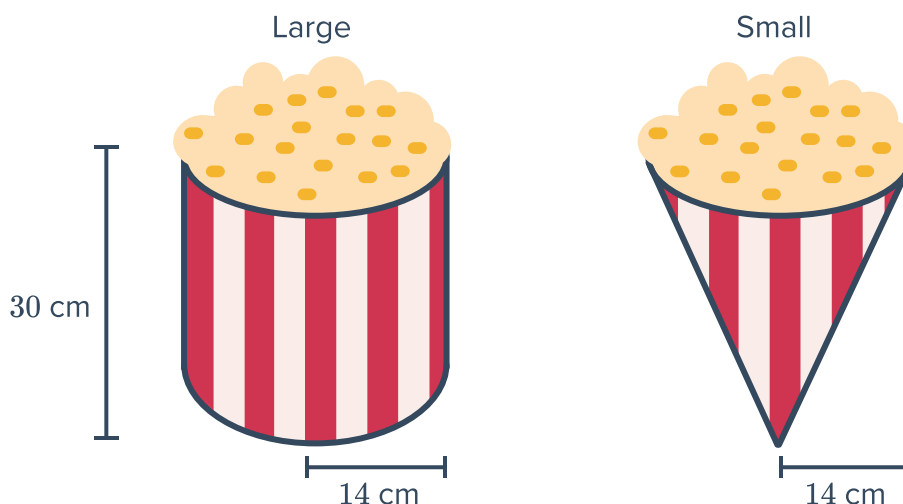


- 20** Consider the following bottles of portable hand wash. Assume that their volumes are equivalent to pyramids of the same dimensions.

- a** By approximately what factor is the amount of hand wash in bottle B greater than in bottle A?
- b** If bottle A costs \$50.88 and bottle B costs \$60.00, which is the better buy?



- 21 A theatre serves popcorn in two different containers. The Small size popcorn comes in a cone-shaped container, and the Large size popcorn comes in a cylindrical container as shown below. Both containers have a radius of 14 cm and a height of 30 cm.



- How many small containers must be purchased in order to have the same amount of popcorn as in one large container?
- Find the volume of the small container. Round your answer to the nearest cubic centimetre.
- Find the volume of the large container. Round your answer to the nearest cubic centimetre.
- The cost of a small container of popcorn is \$1.30 and the cost of a large container of popcorn is \$3.50. Which is the better buy?